Project Completion Summary

Semantic Segmentation of Brain Tumor from MRI Images

Department of Mechanical & Aerospace Engineering — IIT Hyderabad

Project Title	Semantic Segmentation of Brain Tumor from MRI Images
Domain	Deep Learning / Medical Image Analysis
Tools & Frameworks	Python, TensorFlow, Keras
Architecture	U-Net (Encoder–Decoder with Skip Connections)
Dataset	Medical MRI Image Database
Department	Mechanical and Aerospace Engineering, IIT Hyderabad

Objective

The primary objective of this project was to develop an automated deep learning-based pipeline for the **semantic segmentation of brain tumors from MRI images**, enabling accurate delineation of tumor regions without the need for manual annotation or intervention. The project demonstrates the applicability of modern convolutional neural network architectures in the domain of medical imaging.

Methodology

A curated MRI image dataset was obtained from a publicly available medical imaging database for training and evaluation purposes. The segmentation model was developed using the **U-Net architecture**, implemented in **Python** with the **TensorFlow** and **Keras** deep learning frameworks.

The U-Net architecture was selected for its proven effectiveness in biomedical image segmentation, owing to its symmetric encoder–decoder structure combined with skip connections that preserve fine spatial details across network layers.

The workflow followed the steps outlined below:

Step 1. Data Acquisition: MRI images along with corresponding ground truth segmentation masks were downloaded from the database.

Step 2. Preprocessing: Images were resized, normalized, and augmented to improve model generalisation.

Step 3. Model Development: A U-Net model was constructed and trained using TensorFlow/Keras with appropriate loss functions for segmentation tasks.

Step 4. Inference: The trained model was applied to unseen MRI images to generate predicted segmentation masks.

Step 5. Evaluation: Predicted masks were compared against ground truth annotations to assess accuracy.

Results and Outcomes

The trained model was applied to test MRI images, and tumor regions were **successfully segmented** from the input data. The predicted segmentation masks were systematically compared against the corresponding ground truth masks to evaluate model accuracy and reliability. The results demonstrated that the U-Net model was capable of identifying and isolating tumor boundaries with satisfactory precision.

> **Key Result:** Tumor regions were accurately segmented from MRI input images, with the predicted outputs showing strong spatial agreement with the provided ground truth masks.

Key Learnings

Through this project, the students gained hands-on experience in the following areas:

- Medical image processing, dataset handling, and preprocessing pipelines
- Deep learning model design using the U-Net architecture
- Implementation of end-to-end segmentation workflows in TensorFlow and Keras
- Qualitative and quantitative evaluation of predicted outputs against ground truth annotations
- Practical understanding of encoder–decoder networks and skip connection mechanisms

Conclusion

This project successfully demonstrated the application of deep learning-based semantic segmentation for automated tumor detection in MRI images. The U-Net model developed herein provides a strong foundation for future refinement, including the incorporation of larger and more diverse datasets, advanced data augmentation strategies, and rigorous quantitative benchmarking using metrics such as the **Dice Coefficient** and **Intersection over Union (IoU)**.

The experience gained through this project equips the students with essential skills in applied machine learning, medical image analysis, and scientific programming — highly relevant to both academia and industry.

Students

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